Software Validation and Evolution

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Software Validation and Evolution (A36121)

Short Title:	SW Validation and Evolution		
Faculty	Science and Computing		
Department	Computing and Mathematics		
Cluster	Information Systems and Modelling		
Author	BMCKEOWN		
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module aims to provide students with fundamental knowledge and skills related to software validation and verification, including security.

Programmes

`COMP-0082`	Higher Diploma in Science in Business Systems Analysis (`WD_KBUSY_G`)	2 / 3 / M

Indicative Content

- Validation planning
- Testing fundamentals, including test plan creation and test cases
- Black-box and white-box testing techniques, reviews, and static analysis techniques
- Automated testing
- Security testing (identify security needs and risks, including security requirement to the test plan, test for security including strong encrypted password)
- JavaScript programming (variables, functions, conditional statements, and loops
- Unit, integration, validation, performance, regression, and system testing
- Object-oriented testing
- Inspections
- Software mainteneance, reuse and versioning

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Compare and explain testing approaches based on their types, methods and levels.
2. Compare and explain the validation and verification processes.
3. Produce a test plan for a small- or medium-size project.
4. Design and perform manual and automated tests using relevant tools.

Learning and Teaching Methods

- This module will be presented by a combination of lectures and computer-based practicals.
- The lectures will be used to introduce new topics and their related concepts.
- The practicals will be used so that students put their knowledge into parctice to either design or implement test plans and test cases.
- For online delivery the lectures and practicals will be a combination of comprehensive rich media instructional content (notes), interactive synchronous video (live webinars/classes) and asynchronous interactive video playback (on-demand).

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	25%	4
Assignment	25%	1,3
In-Class Assessment	50%	1,2

Assessment Criteria

<40%: Unable to interpret and describe key concepts of the specific knowledge domain(s).
40%-49%: Be able to interpret and describe key concepts of the specific knowledge domain(s).
50%-59%: Ability to discuss key concepts of the specific knowledge domain and ability to discover and integrate related knowledge in other knowledge domains.
60%-69%: Be able to solve problems within the specific knowledge domain(s) by experimenting with the appropriate skills and tools.
70%-100%: All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Tutorial	24	24
Lab	24	24
Independent Learning	87	87

Supplementary Material

- Da Costa, L. _Testing JavaScript Applications_. New York: Manning Publications, 2021.
- Rakitin, S. _Software Verification and Validation for Practitioners and Managers, 2 Edition_. 2nd. MA, USA: Artech House Print on Demand, 2001.
- Saleh, H. _JavaScript Unit testing_. Mumbai, India: Packt Publishing, 2013.

Requested Resources

- COMPUTER LAB: BYOD Lab